

Nutritional Quality Trait Data Collection Protocol

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Purpose: Build a data library of nutritional quality information (lipid, protein, energy density content) for pelagic species in the California Current, as well as for albacore tuna prey species using meta-analysis of published and gray literature. A data collection spreadsheet is included in the database files (Table 11).

Additional documents:

- *11_nutritional_quality_data_collection.csv*
- *11metadata_nutritional_quality_data_collection.csv*
- *Pelagic_Species_Trait_Database.Rmd*

Keyword search procedure & criteria:

1. Using Google Scholar, search the species of interest (“*Genus species*”) with each prey quality metric keyword, as well as (optional) location keywords.
 - a. Conduct searches with the software “Publish or Perish” (see instructions below) to save and easily navigate search results.
 - b. Words in “quotations” are searched with the exact phrase. Without quotations the search engine also uses synonyms (e.g. percent also includes results for proportion).

Quality metric	Keywords	Location keywords (optional)
Lipid Content	“Genus species” “lipid” percent	
	“Genus species” lipid percent	“California”, “Oregon”,
	“Genus species” “fat” percent	“Washington”, “Baja”,
	“Genus species” fat percent	“British Columbia”,
Protein Content	“Genus species” “protein” percent	“Alaska”, “Bering Sea”,
Energy Density	“Genus species” “energy” density	“Japan”, “Kuroshio”,
	“Genus species” “energy” content	“Pacific”
	“Genus species” “calorie” content	
Lipid & Protein	“Genus species” “proximate”	

composition

2. Searches can include 10s, 100s, or 1000s of results based on the keywords. Efficiently sort through results with the following tips:
 - a. Search results include excerpts from the paper that contain the keywords. Use this and the title to decide if a paper is relevant for the data we are trying to collect.
 - b. Review papers can be especially helpful for data collection.
 - c. Examples of phrases from highly relevant papers: ‘proximate composition’, ‘percent lipid’
 - d. Examples of phrases from non-relevant papers: ‘lipid utilization’, ‘lipid solubility’, ‘protein gene coding’, ‘percent’ (used as a statistical term).
 - e. Avoid papers with keywords only present in the references section. However, references within a paper can be useful for finding more publications with nutritional quality data.
 - f. Avoid quality data reported for a specific organ (e.g. viscera only, reproductive organs, bones, head only, liver).
 - g. Proximate refers to the fractional components of lipid, protein, ash, and moisture in an organism.
3. Aim for at least 3 lipid, protein, or energy content data points per species. If a publication includes multiple data points for a species, include all data.
4. Save papers for data collection with the file name containing the species scientific name, life stage (if known), author, and date (e.g. genus_species_adult_Glaser_et_al_2015).
5. Record relevant quality data, covariates, and file name (*file_name*) in the data spreadsheet (Table 11).

Keyword search with *Publish or Perish* software:

Publish or Perish software interfaces with Google Scholar and other literature search engines to retrieve and analyze citations. Searches provide tabulated results with paper information (e.g. authors, title, year, journal, citations, abstract) that can be exported as a csv to log search results, and track progress looking through papers. Download & install *Publish or Perish*:

<https://harzing.com/resources/publish-or-perish>.

1. Select search engine option “Google Scholar”
2. Enter keyword search terms in the “Keywords” section. Reduce “maximum number of results” to 100. Select “Search”.
 - a. Increase the maximum number of search results if you are having trouble finding papers. However, increasing to 1000 can take up to 20 minutes.
 - b. Change the default maximum number of search results to 100 (or other value). Click the menu bar option “Publish or Perish”, “Preferences”. In the Google section change the value for maximum search results.
3. Click “Save Results”, choose “Results as CSV”, then save with the keyword search terms as the file name (e.g. "Thunnus alalunga" "lipid" content.csv). Leave spaces and

quotation marks in. Papers rank reflects the order of the results based on keyword relevance.

4. Sort through search results, view papers, and log your progress:
 - a. “Abstract” column shows the text excerpts from the paper that includes the keywords searched (similar to using the Google Scholar website). This column, as well as title, are the most useful for finding the most relevant papers for nutritional quality data.
 - Can link to papers from the *Publish or Perish* desktop interface, but the “Abstract” section is not displayed, which is needed for selecting papers. To open papers in the desktop interface, select a row with a paper, right click and select “Open Article in Browser” or “Open Full Text in Browser”.
 - b. “ArticleURL” links to the article on the journal website, “FullTextURL” links to the full text PDF (see notes above). The full text option should link directly to the PDF, but sometimes the link can be broken. The article option is more reliable, but requires an extra step.
 - c. Create columns “Checked” and “Use” to log papers you have checked for quality data (*Checked* = yes), included in the database (*Use* = yes) or not included (*Use* = no) based on paper content.
 - d. Feel free to change the order of the columns for easier viewing and navigation.
5. Starting a new search in *Publish or Perish*:
 - a. Click “New”, then “Google Scholar Search”, OR
 - b. Click an empty row below your previous search row, then select “Google Scholar”
6. One nice feature from the *Publish or Perish* desktop is you can see all the papers that cite the selected paper. To access this select a row with a paper, right click and select “Open Citing Works in Browser” or “Open Citing Works in Publish or Perish”. This is the only information that cannot be accessed from the search results .csv file.

Data Collection:

1. In the paper PDF, quickly search for the data of interest using the *find* command (Ctrl + F) to search for the main keywords (lipid, protein, energy or calorie). For older papers that the find command doesn’t work, scan through the table, figures, methods, and results, as these are the sections of the paper the data is likely to be found.
 - a. If no relevant data is found, record *Use* = no on the .csv with search results.
 - b. Log any papers without relevant data that can be referenced to save time if the paper shows up in future searches. When you are done using a keyword search result’s .csv file, copy the information on papers with *Use* = no to a spreadsheet reporting authors, title, year, source, article URL, query date, and any comments.

2. Record Lipid, protein, and energy content data in the data collection spreadsheet (Table 11) to the highest level of detail reported in the paper. Research articles typically report data in (i) wet weight (ww), (ii) dry weight (dw), and/or (iii) ash-free dry weight (afdww). Record data in the appropriate weight type column for a quality variable (e.g. lipid_ww, lipid_dw, lipid_afdww)
 - a. Wet weight is preferred, but we can convert dw and afdww to ww as long as moisture (%; for dw & afdww) and ash (%; for afdww) content is included.
 - i. dw is unusable without moisture content, and afdww is unusable without both moisture and ash content. Data is also unusable if the weight type (e.g. wet, dry, ash-free dry) is not included in the paper.
 - ii. Record unusable papers due to lack of conversion values or details in the log of papers without relevant quality data; include the reason in the *comments*.
 - b. Units: Percent (%) weight for lipid and protein ww, dw, or afdww. Energy density is recorded in kJ/g. This is also listed in the metadata (Table 11). If the data is recorded in different units, convert before recording the data. Note conversion in the comments section.
 - c. Do not use quality data for a specific organ (e.g. viscera only, reproductive organs, bones, head only, liver).
3. Record each data point in the paper for the species of interest in a different row. If the value is a mean of multiple individuals, record the sample size (n) in the database.
4. Record the following covariate information associated with the nutritional data, if available (below).
 - a. Sampling date: If full date (dd/mm/yyyy) is listed, record start_date and end_date. We can use this to get month and year. Only fill out month, year if the full date is not listed. If collection is from a single day, month, or year, put the same value for start and end columns.
 - b. Sampling location: If possible report lat, long, and details on the geographic location. Ocean basin information might need to be inferred from the location.

Covariates collected	
Sample size (n)	Sex
Moisture (%)	Dates collected
Ash (%)	Latitude (decimal degrees)
Length (min-mean-max)	Longitude (decimal degrees)
Weight (min-mean-max)	Geographic location
Life stage	Ocean basin
Age	Reference information

5. Enter “NA” for any values not reported in the paper.
6. If the paper contains quality data for other species, collect data for other species.